Title: Balancing Heritage and Health - Occupational Adaptation in the HAENYEO Community

Abstract

The Haenyeo, traditional female free divers of Jeju Island, South Korea, represent a unique occupational community sustained through shared learning, long-term commitment, and collective responsibility. This case study examines how workforce continuity, skill transmission, and health-related challenges affect the long-term functioning of the Haenyeo system. Recognized by UNESCO as an Intangible Cultural Heritage of Humanity, the Haenyeo tradition depends on intergenerational learning, physical endurance, and strong occupational identity.

Skills are acquired through observation, guided practice, and gradual exposure to increasingly demanding tasks. Senior members play a central role in teaching safety practices, environmental judgment, and diving techniques. However, the workforce is rapidly aging, with limited entry of younger members. This trend reflects increased awareness of long-term health risks and changing career priorities among younger generations.

Rather than viewing this decline only as a problem, this case adopts a health-centered perspective. The shift away from physically damaging work represents an adaptive response focused on well-being and sustainable participation. The case highlights how balancing experience, health, and knowledge preservation is essential for maintaining both cultural identity and safe work practices. The study emphasizes the need for supportive systems that allow cultural knowledge to continue without requiring long-term physical harm to future generations.

Case Scenario

The Haenyeo community operates as a close-knit group of women engaged in traditional free diving for marine harvesting. Entry into this occupation typically occurs through family and community connections, and learning is based on long-term observation and hands-on practice. Younger participants initially engage in shallow and less risky dives, gradually progressing to deeper and more complex tasks under the guidance of experienced members.

Daily work requires repeated breath-hold dives in cold water, exposing workers to significant physical strain. Despite these demands, many divers continue working into old age due to strong attachment to their occupation, financial necessity, and commitment to community traditions. Experienced divers not only contribute to production but also provide guidance, monitor safety, and support decision-making during work activities.

Knowledge related to safe diving practices, marine conditions, and equipment use is transferred mainly through shared experience, storytelling, and direct instruction. While this system has been effective for generations, it is vulnerable due to declining participation by younger individuals. Environmental changes and reduced marine resources have increased the effort required for harvesting, further intensifying physical demands.

As machines and modern harvesting technologies become more common, reliance on physically demanding human labor is reduced. This technological shift can improve safety and efficiency but also changes the traditional role of divers. The gap between experienced workers and new entrants continues to widen, increasing the risk of knowledge loss and placing additional strain on aging workers. This case reflects how traditional work systems must adapt to health, technological, and demographic changes to remain sustainable.

Problem Statement

The main issue in this case is the imbalance between experienced workers and new entrants within the Haenyeo community. The workforce is largely composed of older women, while younger participation is steadily declining. This creates challenges related to physical capacity, safety, and long-term sustainability of the occupation.

The physically demanding nature of the work poses significant long-term health risks, including musculoskeletal strain, respiratory stress, and exposure to cold environments. As awareness of these risks increases, fewer young individuals are willing to enter the occupation. This decline reflects rational, health-based decision-making rather than a lack of respect for tradition.

At the same time, reliance on aging workers increases vulnerability to injury, fatigue, and reduced productivity. The limited number of younger participants also reduces opportunities for gradual learning and shared workload. This places pressure on senior members to continue working beyond safe physical limits.

The situation raises concerns about continuity of skills, safety standards, and the long-term viability of this traditional work system. Without adaptation, the community faces the risk of declining participation, increased health-related problems, and gradual loss of accumulated knowledge.

Causes of Problems

One major cause of declining participation is increased awareness of long-term health consequences associated with repetitive diving, cold-water exposure, and physical strain. Younger generations are more likely to prioritize personal health, safety, and future quality of life when making career decisions.

Social and economic changes have expanded access to education and alternative employment opportunities that are less physically demanding and more stable. This reduces the attractiveness of traditional diving as a long-term occupation.

Environmental factors also contribute to the problem. Climate change, pollution, and overfishing have reduced marine resources, increasing the effort required to achieve the same output. This makes the work more exhausting and less rewarding, particularly for older workers.

The introduction of machines and modern harvesting technologies further reduces dependence on manual free diving. While technology can improve safety and efficiency, it also decreases the need for traditional human labour. Cultural shifts also play a role, as physically demanding traditional occupations may be viewed as less desirable compared to modern professions. Together, these factors accelerate workforce decline and increase pressure on remaining workers.

Solutions & Consequences

Rather than encouraging younger individuals to enter physically harmful work, solutions should focus on adapting the system to protect health while preserving knowledge. One important approach is to create structured opportunities for experienced divers to transition into guidance, teaching, and safety-support roles. This allows older workers to remain involved while reducing physical strain.

Technology and machines can be used to handle the most physically demanding aspects of harvesting, while human expertise can focus on monitoring conditions, ensuring quality, and making safety-related decisions. This combination can improve efficiency and reduce health risks.

Knowledge preservation should be strengthened through documentation, training demonstrations, and cultural education programs. Recording techniques, safety judgment, and environmental knowledge can help protect accumulated expertise even as active diving declines.

Community and institutional support can promote heritage-based activities such as cultural tourism, museums, and educational outreach. These alternatives allow cultural identity and economic value to continue without requiring long-term physical harm.

The consequences of these adaptive strategies would include a smaller but safer workforce, reduced health burden on older workers, and continued transmission of cultural knowledge. Failure to implement such changes may result in further decline, increased injury risk, and eventual disappearance of the physically demanding form of Haenyeo work.

Research Evidence & Literature Review:

Multiple studies and official reports confirm a sharp decline and aging trend in the Haenyeo workforce. Recent research published in *Psychiatry Investigation* (2025) reports that there are approximately **2,600–2,800 active Haenyeo**, and **over 90% are aged 60 years or older**. This indicates a severe imbalance between older and younger participants and highlights limited workforce renewal.

Earlier longitudinal data show a dramatic population decrease over time. Lee (2017) reported that the number of Haenyeo declined from **over 14,000 in 1970 to around 4,000 in 2015**, with the average age rising significantly. The same study noted that nearly **88% of Haenyeo were over 60 year's old**, confirming rapid workforce aging.

UNESCO (2023) further reports that current estimates range between **3,200 and 3,300 divers**, compared to **15,000–20,000 in earlier decades**. This confirms that the traditional workforce is shrinking despite heritage preservation efforts.

Health-focused studies also show high levels of chronic illness, hearing problems, musculoskeletal disorders, and injury risk among older Haenyeo. Kim et al. (2015) found that the **average age was over 74 years**, with a high prevalence of chronic disease and injury-related hospital visits. These findings support the argument that continued full-scale diving poses serious long-term health risks.

Overall, the literature strongly supports the conclusion that the decline in participation is linked to aging, health concerns, environmental stress, and socio-economic change. These findings justify a health-centered and adaptive approach rather than blaming younger generations for reduced participation.

Decline in Number of Haenyeo Divers (1970–2025)

Year	Approx. Number of Haenyeo
1970	14,000+
2015	4,000
2020	3,200
2024	2,800
2025	2,600

(Source: Lee, 2017; UNESCO, 2023; Park et al., 2025)

SWOT Analysis:

Strengths:

- Strong experience-based knowledge
- High occupational commitment
- Cooperative work culture
- Global cultural recognition

Weaknesses:

- Aging workforce
- High physical demands
- Informal knowledge transfer

Opportunities:

- Teaching and mentoring roles
- Cultural and educational programs
- Technology-assisted work
- Institutional and heritage support

Threats:

- Environmental degradation
- Declining youth participation
- Over-reliance on aging workers
- Loss of traditional skills

Conclusion:

The future of the Haenyeo is likely to involve significant transformation rather than full continuation of traditional practices. Health-centered decision-making, technological advancement and changing work preferences make it fair and reasonable for younger generations to avoid physically damaging occupations. The tradition is more likely to survive through adaptation, where cultural knowledge and identity are preserved in safer, less physically harmful forms. This represents a sustainable and humane approach that prioritizes well-being while maintaining cultural continuity.

References

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